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6th OPJU International Technology Conference (OTCON 6.0) on SMARTICON (Smart & Intelligent Computing for Next-Generation Technologies) : Submission (331) has been created.

1 message

Microsoft CMT <noreply@msr-cmt.org> Sun, Nov 1, 2026 at 8:33 PM
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Hello,

The following submission has been created.

Track Name: Smart Computing for IoT, Edge Computing & Intelligent Sensor Systems

Paper ID: 331

Paper Title: CHROMA: Colour Intelligence-Bridging Physical Colours and Digital Design

Abstract:
Transferring colours from physical surfaces into digital design applications is complicated by illumination, surface reflectance, sensor response and fragmented software workflows. This paper presents CHROMA, a prototype integrating CHROMAPOD, a portable ESP32–TCS34725 colour scanner, with CHROMA Studio, a mobile-compatible colour analysis application. CHROMAPOD supports acquisition, local CIE76 comparison and Bluetooth Low Energy (BLE) exchange; the companion application provides format conversion, CIEDE2000 image matching, surface recolouring and colour-reference management. Calibration was evaluated on 16 printed RGB reference patches, with five complete scans per patch for each of two firmware versions (80 scans per version). Mean channel-wise RGB agreement against intended digital codes increased from 85.02% in V13 to 94.68% in V14, a 9.66 percentage-point improvement. These values quantify agreement with nominal digital references under the experimental conditions, not certified colourimetric accuracy, because the printed swatches were not independently characterized and some reference colours informed calibration development. The results demonstrate a physical–digital workflow requiring independent validation.

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Secondary Subject Areas: Not Entered

Submission Files:
CHROMA Colour Intelligence_Bridging Physical Colours and Digital Design.pdf (734 Kb, Sun, 01 Nov 2026 15:01:36 GMT)

Submission Questions Response: Not Entered

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